Phuc Nguyen Tran Thien

AI & Machine Learning · Computer Vision · Research & Engineering

Ho Chi Minh City, Vietnam · phucntt.258i@ou.edu.vn

Portfolio / GitHub / LinkedIn / Google Scholar / ORCID

AI practitioner with experience in computer vision, multimodal learning, OCR and semantic search. Build training/evaluation pipelines, inference tools and application APIs. Research Assistant and M.Sc. student; co-author of six publications. Open to AI/ML engineering roles, research assistantships and international graduate opportunities.

TECHNICAL SKILLS

AI/ML: Python, PyTorch, scikit-learn, OpenCV, MediaPipe, ONNX Runtime; model training, dataset audits, error analysis and reproducible evaluation.
Engineering: FastAPI, REST APIs, WebSocket, Docker, Git, Linux, Flutter.
Retrieval: OCR, Sentence Transformers, Elasticsearch, MongoDB, BM25 and dense retrieval.

EXPERIENCE

Research Assistant · Ho Chi Minh City Open University Aug 2025-Present

• Develop computer vision and multimodal models for visual speech, Vietnamese isolated signs and smartphone chemical sensing.

• Build PyTorch experiments, inspect dataset splits and model errors, and develop reusable inference tools and application prototypes.

• Co-author six publications, including two first-author IEEE JCSSE papers and an article in *Analytica Chimica Acta*.

AI Engineer Intern · FPT Software Sep-Dec 2024

• Built OCR and semantic-search components using EasyOCR, Sentence Transformers, Elasticsearch and MongoDB.

• Integrated hybrid BM25/dense retrieval, metadata filtering and REST APIs with application teams.

SELECTED AI PROJECTS

AI-Chemistry - Developed colorimetric-strip preprocessing, multi-task concentration prediction and a FastAPI inference service. Connects model experiments with an application-facing API.

LipTrace - Built dataset checks, CTC model bundles, ONNX export and corpus-level PyTorch/ONNX comparison, with explicit input and model-configuration checks.

VSL Streaming Toolkit - Developed a Python library and WebSocket service for ordered keypoints, isolated-sign segmentation, model-input validation and per-connection state.

Vietnamese Lip Reader - Built a Flutter + FastAPI application with video upload, a bounded inference job queue, transcript review and subtitle workflows.

EDUCATION

Ho Chi Minh City Open University Aug 2025-Present

M.Sc. in Computer Science (Artificial Intelligence), in progress; expected 2027.

FPT University, Ho Chi Minh City 2021-2025

Bachelor's degree in Artificial Intelligence.

RESEARCH AND PUBLICATIONS

Focus: visual speech recognition, Vietnamese sign language, multimodal learning and AI for scientific sensing. Six published papers; two first-author papers.

1. **Phuc Nguyen Tran Thien**, Trung Nguyen Quoc, Thai-Binh Tran, Vinh Truong Hoang, Kiet Tran Trung (2026). **Deep Smartphone-Based Classification of NH4+ and NH2- Colorimetric Strips: Revisiting the Role of Explicit Green-Border Calibration**.
IEEE JCSSE 2026 · First author. doi:10.1109/jcsse68839.2026.11597064

2. **Phuc Nguyen Tran Thien**, Trung Nguyen Quoc, Thanh-Danh Nguyen, Vinh Truong Hoang, Kiet Tran-Trung (2026). **Deep NH4+ Regression for Smartphone Colorimetric Test Strips: On the Need for Explicit Green-Border Normalization**.
IEEE JCSSE 2026 · First author. doi:10.1109/jcsse68839.2026.11597039

3. Trung Nguyen Quoc, **Phuc Nguyen Tran Thien**, Vinh Truong Hoang, Kiet Tran-Trung (2026). **Adapting Hierarchical Feature Aggregation and Spatio-Temporal Graph Neural Networks for Vietnamese Visual Speech Recognition on a Small Phrase-Level Dataset**.
IEEE CICN 2026. doi:10.1109/cicn70047.2026.11594397

4. Trung Nguyen Quoc, **Phuc Nguyen Tran Thien**, Vinh Truong Hoang, Kiet Tran-Trung (2026). **Toward Edge-Deployable Vietnamese Lip Reading: A Compact Lipnet-Style Pytorch Recognizer with a Jetson-Oriented Inference Pipeline**.
IEEE CICN 2026. doi:10.1109/cicn70047.2026.11594266

5. Trung Nguyen Quoc, Minh-Vuong Phan, Khoi Pham Dang, Huy Ngo Huu Gia, **Phuc Nguyen Tran Thien**, et al. (2026). **Smartphone-based colorimetric sensing with reference calibration and ensemble machine learning for enhanced detection of nitrite and ammonium ions**.
Analytica Chimica Acta, 1388, 345097. doi:10.1016/j.aca.2026.345097

6. Trương Duy Việt, Ngô Hữu Gia Huy, Phạm Đăng Khôi, **Nguyễn Trần Thiên Phúc** (2025). **MoViNet-A2 cho bài toán nhận diện ký hiệu Tiếng Việt**.
HCMCOU Journal of Science - Engineering and Technology, 20(2), 3–14.
doi:10.46223/hcmcoujs.tech.vi.20.2.4201.2025

CURRENT RESEARCH MANUSCRIPTS

Visual context and text scope in visual speech recognition - research manuscript, public code and numerical reproduction.
Vietnamese isolated-sign recognition and sign-lip fusion - working manuscript, implementation and result archives.

SELECTED AWARDS

1st Place, Cardano SEA Hackathon (AI Track).
Third Prize, Food Innovation and Development 2025, Food Security & Social Impact category (student team lead).
Top 10, Southeast Asia Sparki Startup Competition; Top 5, Youpreneur Launchpad 2024; Top 10, I-Startup 2024.